A v. S

Apple vs. Samsung

A Strategic Economic Analysis

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COURT
SAMSUNG
APPLE
SAMSUNG
3G
68°

Apple won in court.
Samsung won the market.

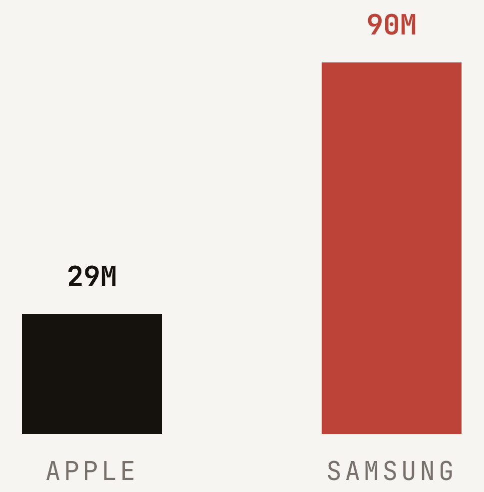
How can both be true?

AUGUST 2012 — VERDICT

$1.05B

LARGEST PATENT VERDICT IN U.S. HISTORY

Q2 2012 SMARTPHONE UNITS SHIPPED



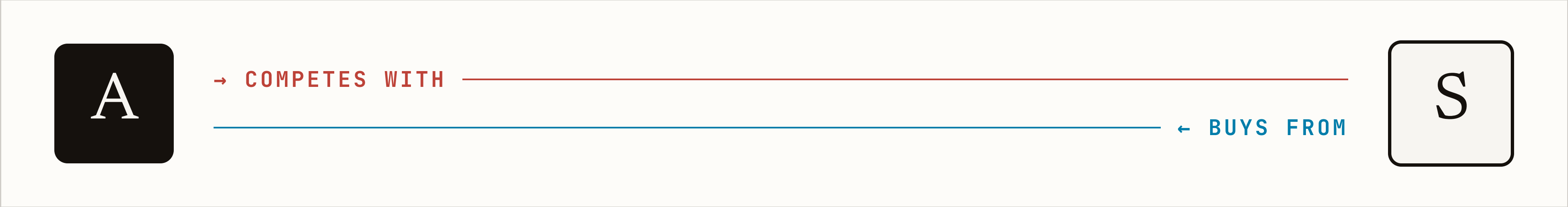
FROM QUESTION TO VERDICT

Our argument in *six beats*.

I · ONE STRANGE RELATIONSHIP

Rivals *and* supplier–customer at the same time.

This duality — **coopetition** — constrains every move that follows in the dispute.



FACT	NUMBER
Samsung's share of iPhone manufacturing cost	26%
Apple's component purchases from Samsung (2010)	$6B
Apple's share of Samsung's total revenue	4%

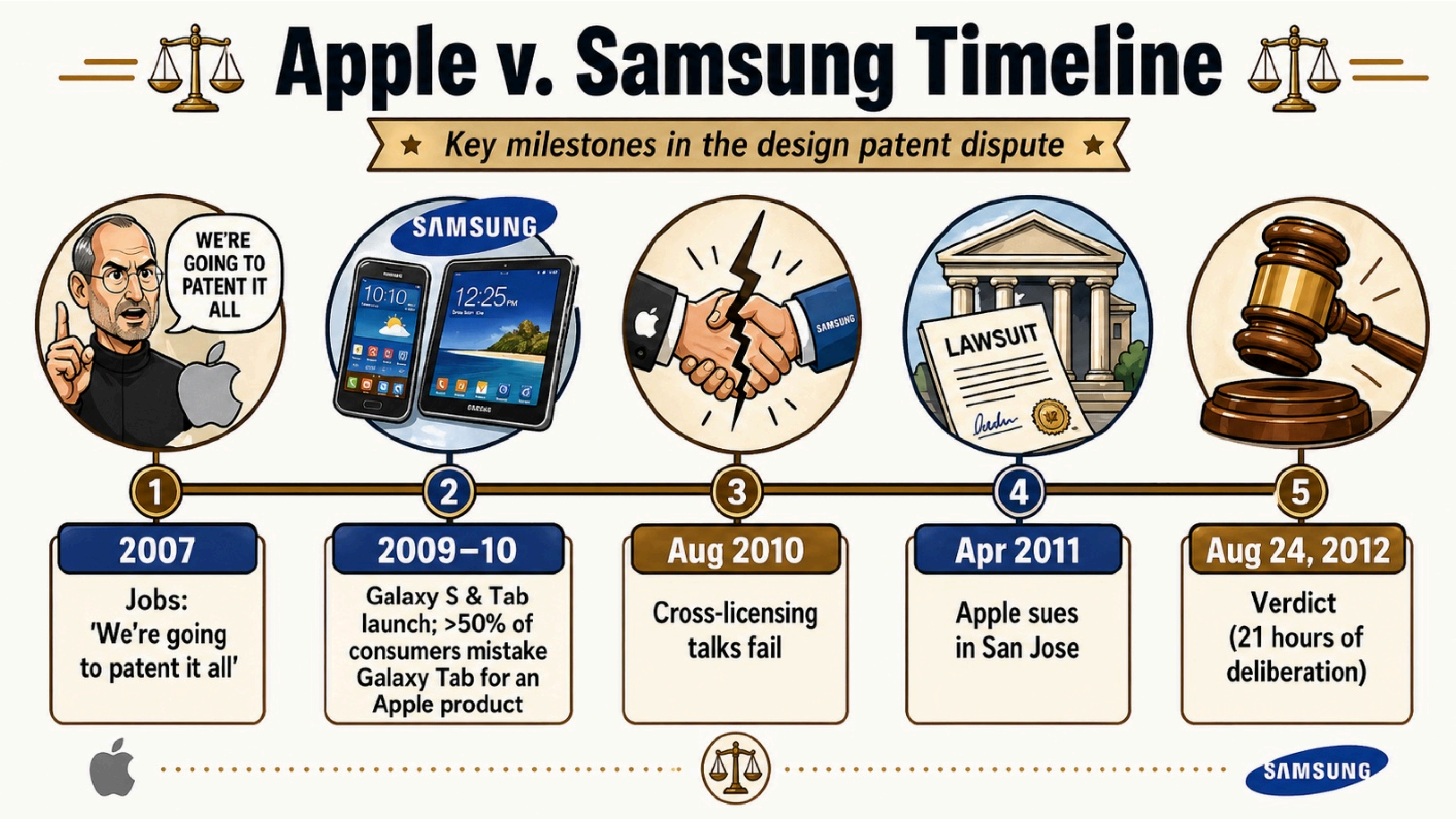
> *"Apple was reluctant to take legal action against Samsung, given its key role in supplying chips and components."*
>
> — WESLEY CASE STUDY

SEVEN YEARS IN FIVE BEATS

From Jobs's promise to the largest patent verdict in U.S. history.

SPAN
2007 → 2012
5 BEATS · 1 VERDICT

APPLE VS. SAMSUNG — PATENT DISPUTE TIMELINE
SOURCE: WESLEY CASE STUDY



II · APPLE DID NOT SUE OUT OF ANGER

Three economic forces pushed Apple *into court*.

Each one is the subject of one of the next three slides.

1 PRESSURE #1

Patent Hold-Up

Too many patents, no clean way out — and the standard fix (cross-licensing) had failed.

2 PRESSURE #2

Lost Differentiation

Samsung looked too much like Apple, and Apple's premium was eroding.

3 PRESSURE #3

Platform Tipping

Android was running away with the smartphone OS market — second place might not survive.

IF YOU REMEMBER NOTHING ELSE FROM THIS DECK — REMEMBER THESE THREE.

WHY APPLE SUED — I

When no one owns the road, everyone pays a toll.

~10K patents needed
TO SHIP ONE SMARTPHONE

$10B / year, industry-wide
ON PATENT LITIGATION BY 2011

> *Apple & Google each spent* ***more on lawsuits than R&D.***
> — ECEB SECTION 3.1

Cross-licensing is the textbook fix. It failed in Aug 2010.

PATENT STRATEGY — COMPARED

	APPLE	SAMSUNG
Offensive	Sue Samsung for copying iPhone	Counter-sue over wireless patents
Defensive	"Patent everything" — wall of patents	Build own patent stockpile
Bargaining chip	Win cases → better licensing terms	Wireless patents as leverage

WHY APPLE SUED — II

Bertrand Competition & Hotelling, in plain sight.

BERTRAND COMPETITION

Identical products → price war → zero profit.

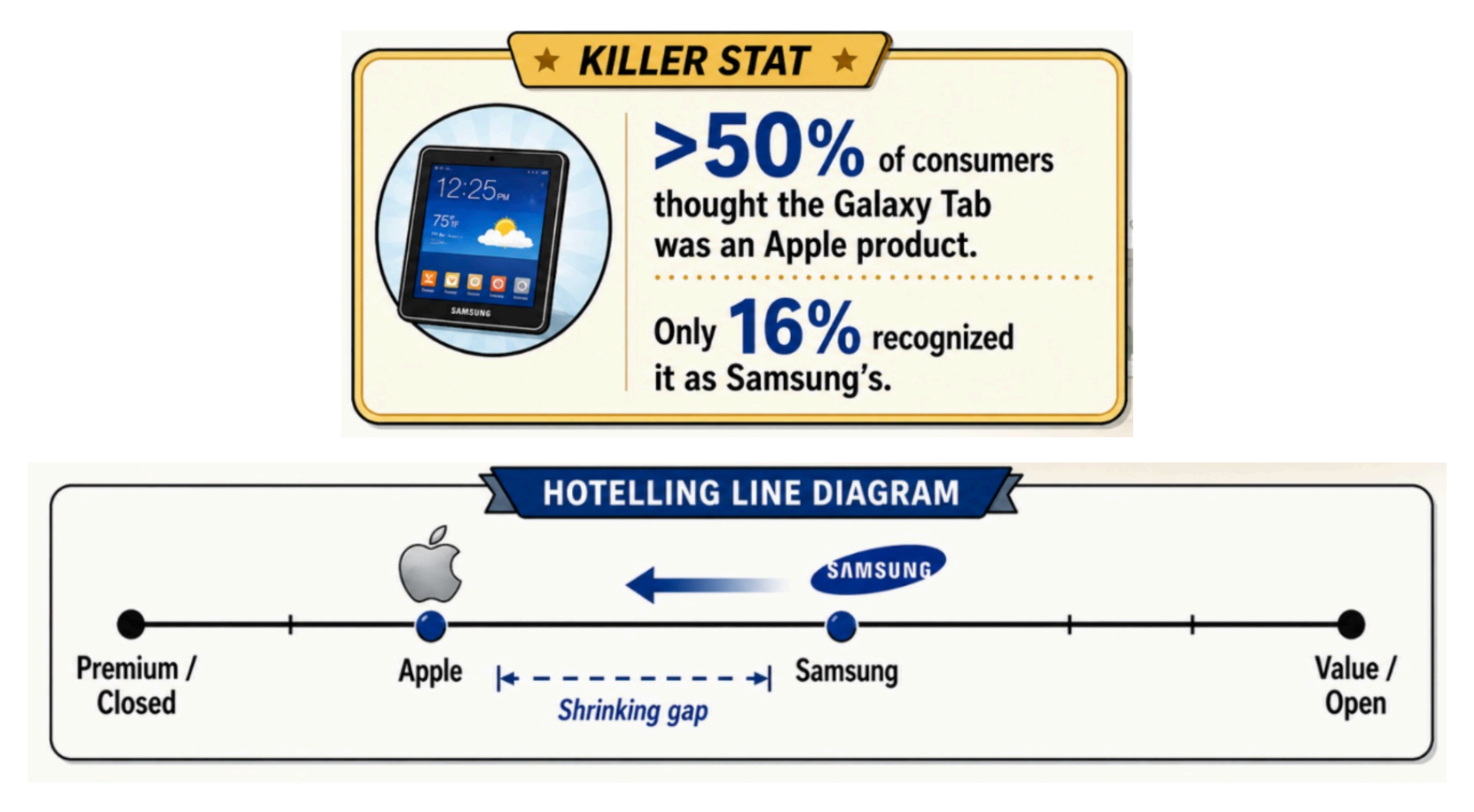
HOTELLING

The closer rivals stand on the "taste line," the harder they compete.

IMPLICATION

Samsung's copying moved it toward Apple → eroded the premium.

FIG. 2 · KILLER STAT + HOTELLING LINE



CAPTION: AS SAMSUNG'S POSITIONING APPROACHED APPLE'S, THE DIFFERENTIATION PREMIUM ERODED.

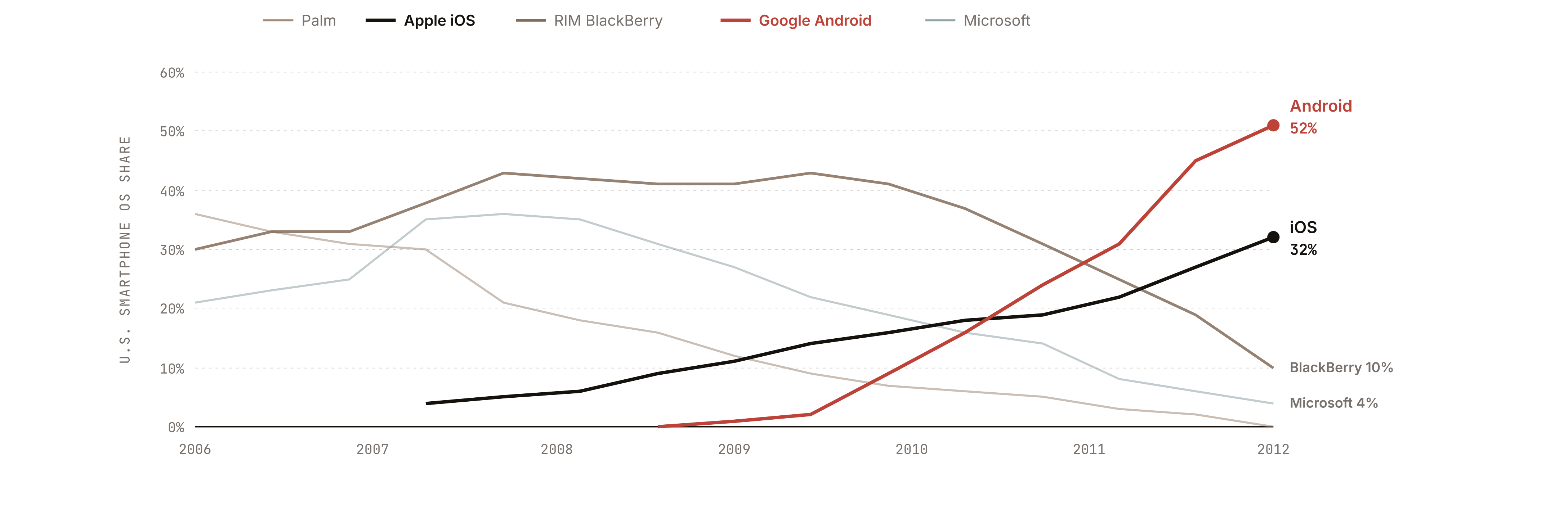
WHY APPLE SUED — III

Network effects don't reward second place.

TWO PLATFORMS, TWO GAMES

	IOS (APPLE)	ANDROID (SAMSUNG+)
Standard	Closed	Open
Network effect	Apps → users	Same — across many OEMs
Switching costs	High (apps, iCloud)	Low between brands
Game	Competes **FOR** the market	Competes **IN** the market

FIG. 3 · U.S. SMARTPHONE MARKET SHARE, 2006-2012



SOURCE: CASE EXHIBIT 10 — CANALYS, SILICON VALLEY INSIDER, COMSCORE, GARTNER.

III · THE PIVOT — APPLE'S THREAT HAD A CEILING

> *"I will spend my last dying breath... every penny of Apple's $40 billion in the bank... I'm going to destroy Android."*
>
> — STEVE JOBS

STRATEGIC COMMITMENT — 3 CRITERIA

Observable? ✓ YES
Public lawsuit + injunctions sought

Understandable? ✓ YES
Copy us → face $1.05B+ costs

Credible? ⚠ PARTIAL
Sunk costs, reputation on the line

WHY ONLY PARTIAL?

PUSH TO SUE

Already suing **HTC**, **Motorola**, **Nokia**, **Amazon** and **Google** (Exhibit 2) → softening on Samsung breaks every other threat.

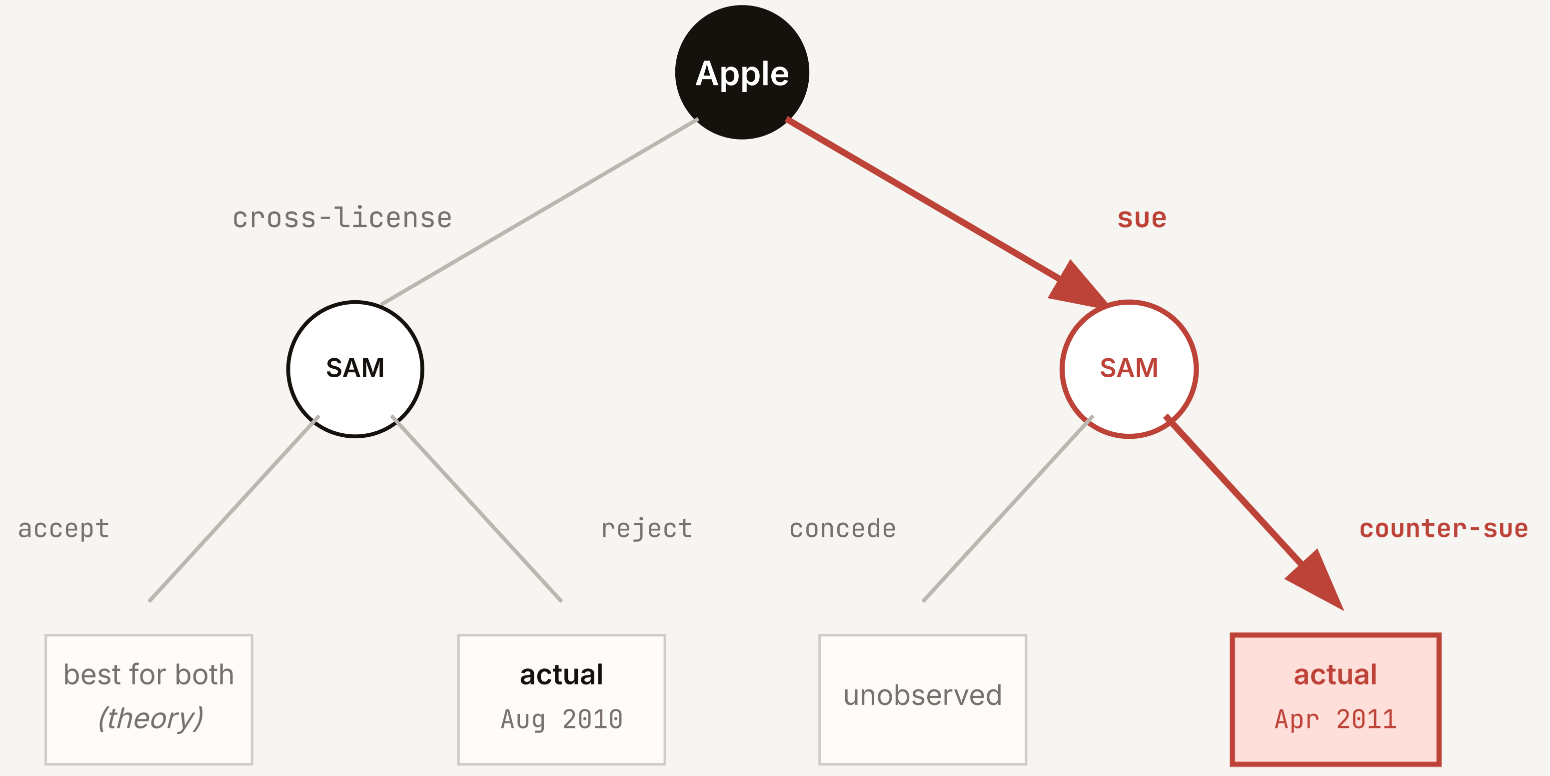
trigger strategy / reputation in a repeated game · suit is the minimum-viable move

PULL TO HOLD BACK

Samsung supplies Apple's chips, screens, memory → can't actually push to ban.

coopetition / hold-up

GAME TREE — HIGHLIGHTED: ACTUAL PATH



CROSS-LICENSE → REJECT → SUE → COUNTER-SUE

APPLE'S THREAT LACKED FULL CREDIBILITY — BOTH SIDES SETTLED IN COURT.

IV · COOPETITION LOGIC

Apple and Samsung play *two games at once* — and the second one saved the relationship.

GAME 1 Smartphone Competition

PRISONER'S DILEMMA

	SAM: DIFFERENTIATE	SAM: IMITATE
APPLE: TOLERATE	Mod π, Mod π	Low π, **High π**
APPLE: SUE	−Legal, 0	−Legal, −Legal

GAME 2 Supply Chain Cooperation

REPEATED GAME

	SAM: RELIABLE SUPPLY	SAM: DISRUPT
APPLE: CONTINUE	★ **$6B+, $6B+** NE	Crippled, Short gain
APPLE: DIVERSIFY	Transition cost, Lose customer	Both lose, Both lose

$$r \leq \frac{\pi^{C} - \pi^{NC}}{\pi^{D} - \pi^{C}}$$

COOPERATION INEQUALITY

r	DISCOUNT RATE	your impatience · ~5–10%
π^{C}	COOPERATE	**$6B** / **$6B** · supply & buy continues
π^{NC}	NO COOPERATION	price war wipes profit · ≈ **$0**
π^{D}	DEFECT ONCE	short-term spike, then relationship collapses

Long-term reward ÷ one-time temptation. With **$6B/yr** at stake, the inequality holds easily.

SAM = Samsung

WHO INNOVATES, AND WHY

Two effects, two players, two strategies.

REPLACEMENT EFFECT

Apple — the entrant.

Nothing to lose.
Pre-iPhone, Apple had **$0** smartphone revenue.

Big disruptive bet.
The touchscreen iPhone — bet the company on a new category.

EFFICIENCY EFFECT

Samsung — the incumbent.

Now has profits to protect.
Incremental gains: better screens, chips, scale.

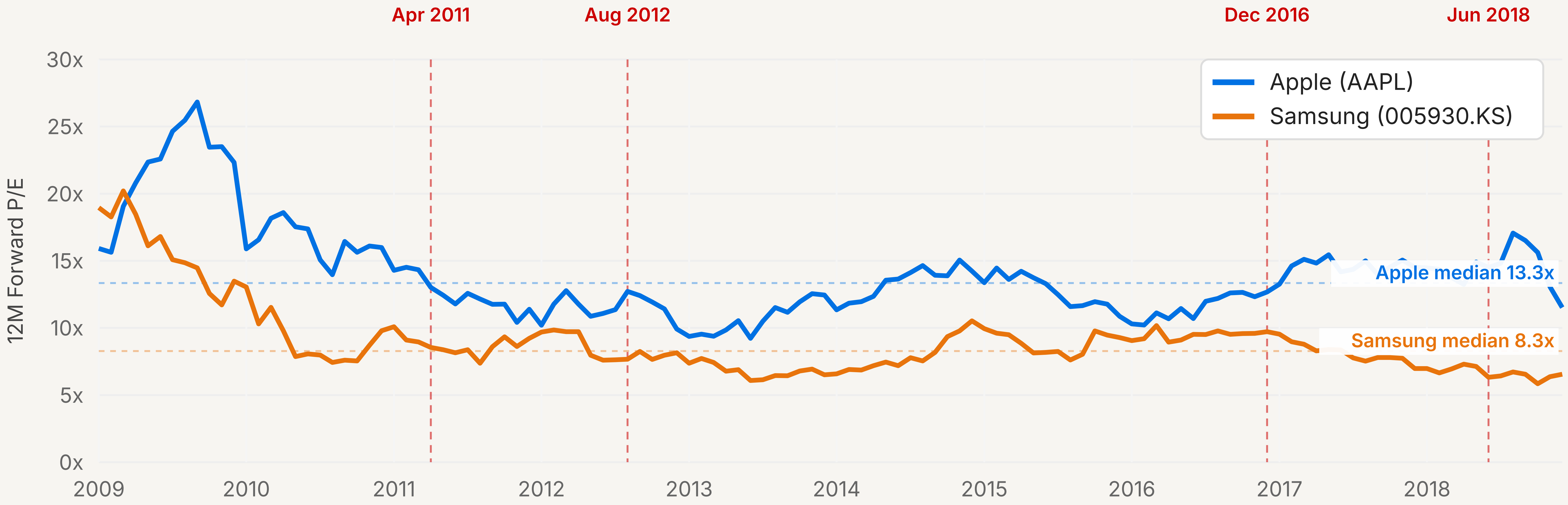
R&D climbs incrementally.
$8.3B → $9.1B (2010 → 2011).

THE PRINCIPLE

*"The more you have to lose,
the smaller your bets."*

V · WHAT MARKETS SAID

Investors never re-rated either firm on lawsuit news.



APR 2011 · SUIT FILED
Apple Δ **−1.7x** · Samsung Δ **−0.4x**

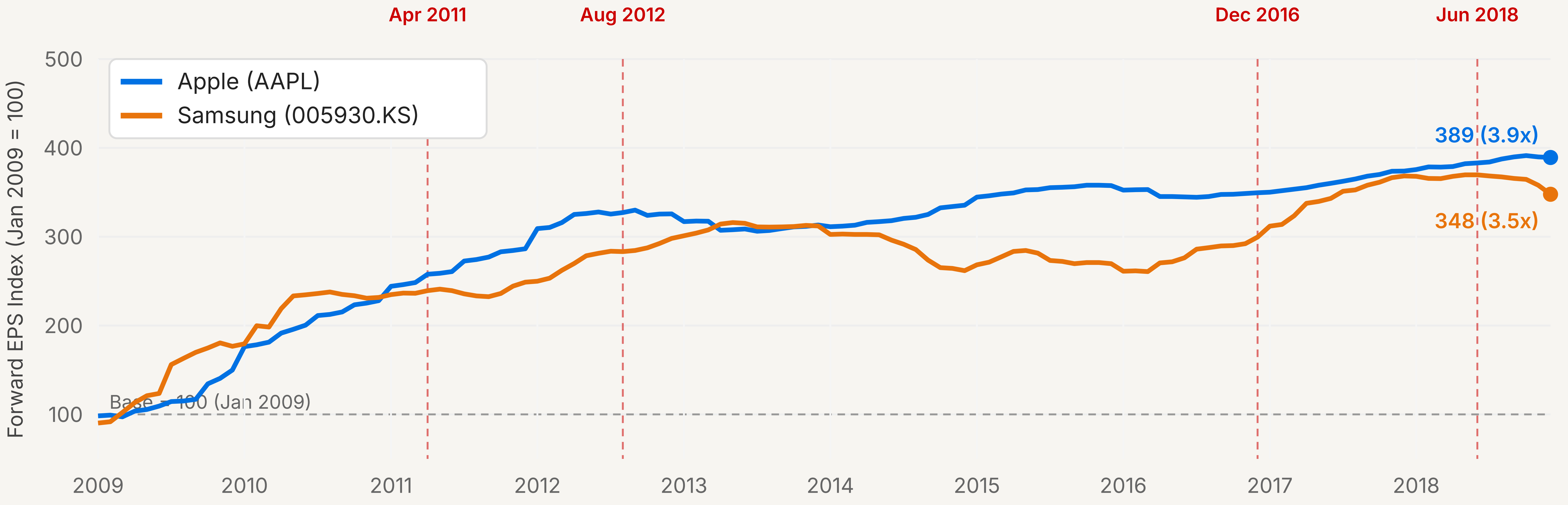
AUG 2012 · VERDICT
Apple Δ **+1.2x** · Samsung Δ **+0.8x**

DEC 2016 · SCOTUS
Apple Δ **+0.6x** · Samsung Δ **+0.1x**

JUN 2018 · SETTLED
Apple Δ **+0.3x** · Samsung Δ **−0.5x**

WHAT EARNINGS DID

Earnings nearly 4× for both firms — through the entire dispute.



APPLE — FORWARD EPS INDEX
100 → 389 (3.89×)

SAMSUNG — FORWARD EPS INDEX
100 → 348 (3.48×)

VI · THE VERDICT, FIVE YEARS LATER

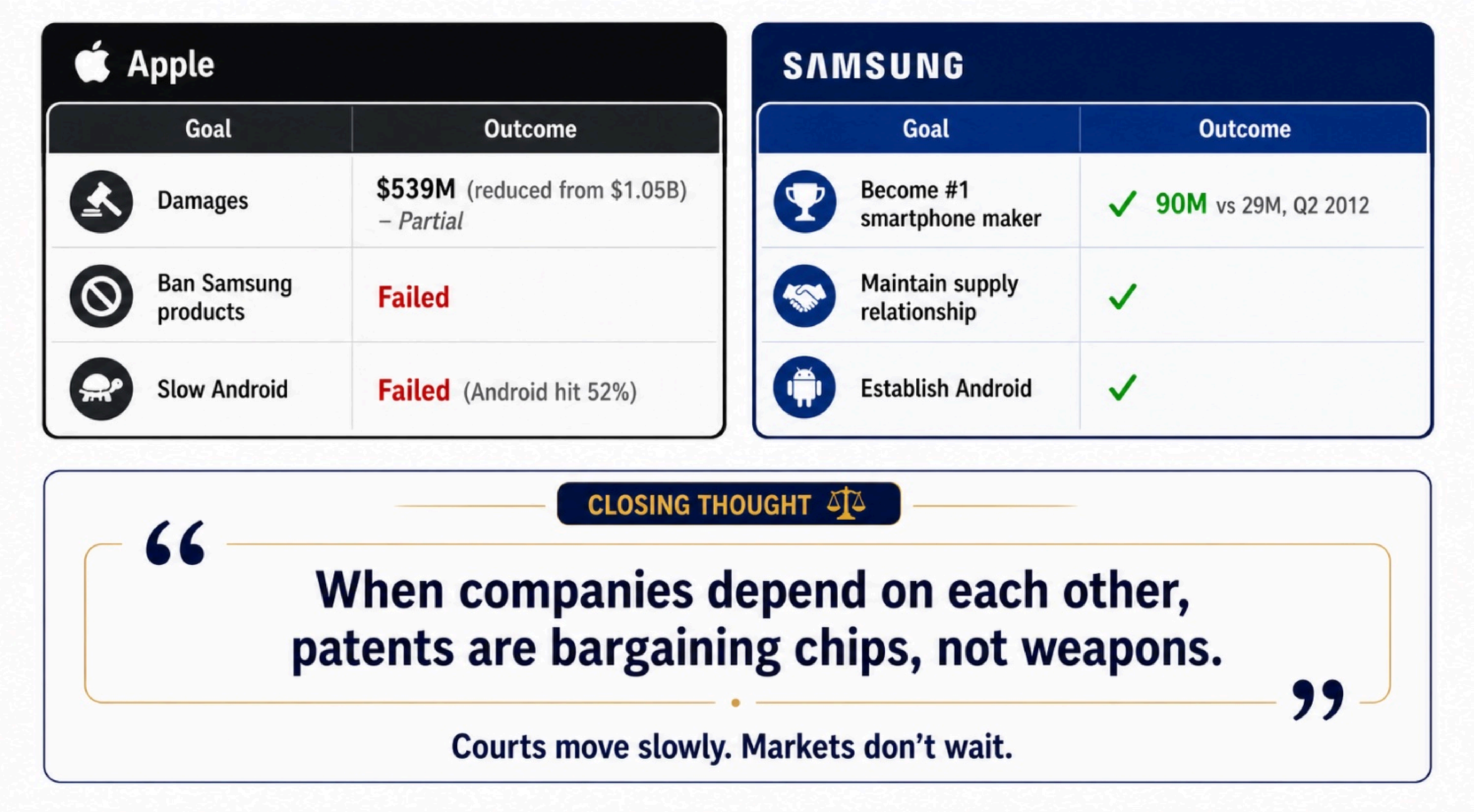
Two scorecards. Two stories.

DAMAGES AWARDED

$539M

8.4% OF SAMSUNG'S 2011 NET INCOME

GOAL OUTCOMES — APPLE VS. SAMSUNG



Apple

Goal	Outcome
Damages	$539M (reduced from $1.05B) – *Partial*
Ban Samsung products	Failed
Slow Android	Failed (Android hit 52%)

SAMSUNG

Goal	Outcome
Become #1 smartphone maker	✓ 90M vs 29M, Q2 2012
Maintain supply relationship	✓
Establish Android	✓

CLOSING THOUGHT

"When companies depend on each other, patents are bargaining chips, not weapons."

Courts move slowly. Markets don't wait.

EPILOGUE I · SAMSUNG'S SCALE

Six years of growth, in USD billions.

METRIC	2006	2007	2008	2009	2010	2011
Sales	89.4	106.0	110.1	106.8	133.8	148.9
R&D	*	*	*	5.8	7.9	9.0
Net Income	8.3	8.0	5.0	7.6	14.0	12.4
Net Margin	9.3%	7.5%	4.6%	7.2%	10.4%	8.3%
Equity	50.2	60.2	57.1	57.2	77.3	91.9

SOURCE: SAMSUNG AUDIT REPORTS. CONVERTED TO USD AT FRB H.10 PERIOD-AVERAGE KRW/USD: 2006 955.5 · 2007 929.3 · 2008 1,102.1 · 2009 1,276.9 · 2010 1,156.1 · 2011 1,108.3. *R&D BUNDLED IN SG&A PRE-2009 (K-GAAP).

EPILOGUE II · FIVE YEARS LATER

Seven years of litigation, quietly settled.

2016

U.S. SUPREME COURT — 8-0

Design-patent damages should be calculated only on the *copied component* (e.g., the front face), not the whole phone — narrowing the $399M design-patent portion.

2018

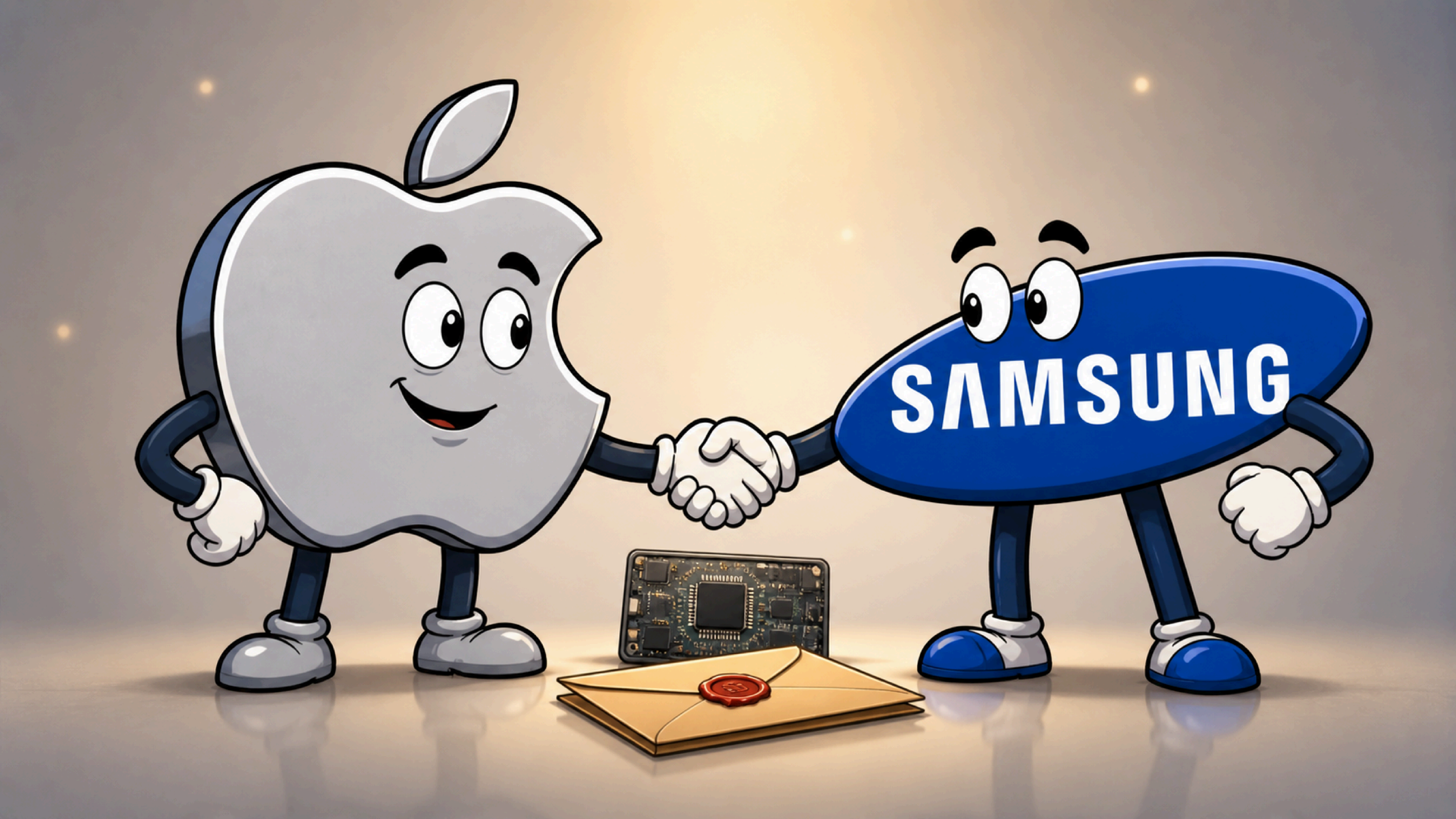
RETRIAL & SETTLEMENT

$539M awarded. A confidential settlement in June 2018 ended seven years of litigation.

& yet

THE QUIET TRUTH

Despite the war, the supply partnership continued.


SAMSUNG